Total No. of Questions : 6]

SEAT No. :

P105

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Oct.-16/B.E./Insem.- 163

B.E. (Information Technology) (Semester - I)

MODERN COMPILERS (Elective - I (c))

(2012 Pattern)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) ***Neat diagrams must be drawn wherever necessary.***
2) ***Assume suitable data, if necessary.***

Q1) a) Write the translation of following statement into IR tree:
if x<5 then a>b else 0. **[4]**

b) Describe each phase of compiler in brief. **[6]**

OR

Q2) a) For each variable a,b,c,d in the given C program, say whether the variable should be kept in memory or a register and why? **[4]**

```
int f(int a, int b)
{     int c[3], d;
      d=a+1;
      b=g(c,&b);
      return c[1]+b;
}
```

b) Define stack frame? Explain the concepts of Stack Pointer (SP) and Frame Pointer (FP) with respect to stack frame. **[6]**

Q3) a) Define canonical trees. Describe any one identity for transformation on ESEQ. **[4]**

b) Write an algorithm for optimal tiling instruction selection and comment on its efficiency. **[6]**

OR

Q4) a) Define flow-graph terminologies with the help of suitable flow-graph. **[4]**

i) Predecessor

ii) Successor

iii) Uses

iv) Defs

b) What is the architecture of Pentium computers? Write any two problems and corresponding solution of this architecture with respect to instruction selection? **[6]**

Q5) a) What is coalescing? Explain any one strategy for the same. **[6]**

b) Explain the concepts of precolored nodes with suitable example. **[4]**

OR

Q6) a) What is incremental garbage collection? Describe tricolor marking. **[4]**

b) Explain the algorithm of Mark-and-Sweep garbage collection. **[6]**

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